

Credit Card Payment System

Full Stack Development Task

Submitted By: Dineshkumar S

Role: Python Full Stack Developer

Task: Credit Card Payment System — Full Stack Assignment

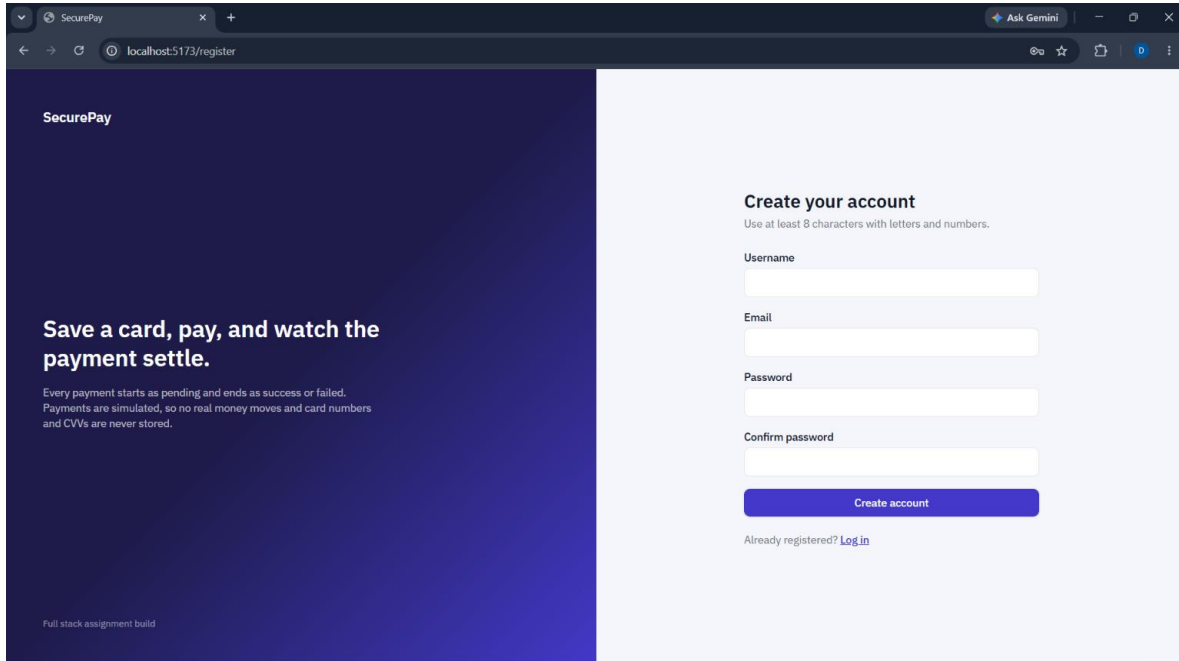
Date: 05/10/2026

Requirement Completion

Requirement	Status	Evidence
User Registration	Done	Register page and API working.
Login with JWT	Done	Access and refresh tokens returned (200).
Logout and Token Blacklist	Done	Refresh tokens blacklisted on logout.
Password Security	Done	Passwords stored as PBKDF2 hashes.
Card Management	Done	Cards saved with masked number and last4 only.
No CVV Storage	Done	cards table has no CVV column.
Simulated Payments	Done	POST /payments returns 201 SUCCESS; no real gateway.
Payment Status	Done	SUCCESS and FAILED results stored and shown.
Payment Validation	Done	Invalid card rejected with 404.
Transaction History and Filters	Done	Filter by status, amount and date.
CSV Export	Done	Export works and is logged in admin_logs.
Admin Panel	Done	Daily summary, users, cards and transactions.
Frontend (React)	Done	Register, login, dashboard and admin pages.
API Documentation	Done	Swagger available for Django and FastAPI.
Django Unit Tests	Done	28 tests passing.
Database	Done	MySQL schema in place; dump exported.
Docker	Done	db, fastapi, django and frontend containers running.
Postman Collection	Done	Collection imported and run; requests return 201 and 200 with tests passing.

Evidence

Authentication — Register Page



The screenshot shows a web browser window with the URL `localhost:5173/register`. The page is titled "SecurePay" and features a dark blue sidebar on the left with the text "Save a card, pay, and watch the payment settle." and a disclaimer: "Every payment starts as pending and ends as success or failed. Payments are simulated, so no real money moves and card numbers and CVVs are never stored." The main content area is light blue and contains the "Create your account" section. It includes a sub-header "Create your account" and a note "Use at least 8 characters with letters and numbers." Below this are four input fields: "Username", "Email", "Password", and "Confirm password". A blue "Create account" button is positioned below the fields. At the bottom of the form, there is a link "Already registered? Log in". The footer of the page reads "Full stack assignment build".

SecurePay

Save a card, pay, and watch the payment settle.

Every payment starts as pending and ends as success or failed.
Payments are simulated, so no real money moves and card numbers
and CVVs are never stored.

Full stack assignment build

Create your account

Use at least 8 characters with letters and numbers.

Username

Email

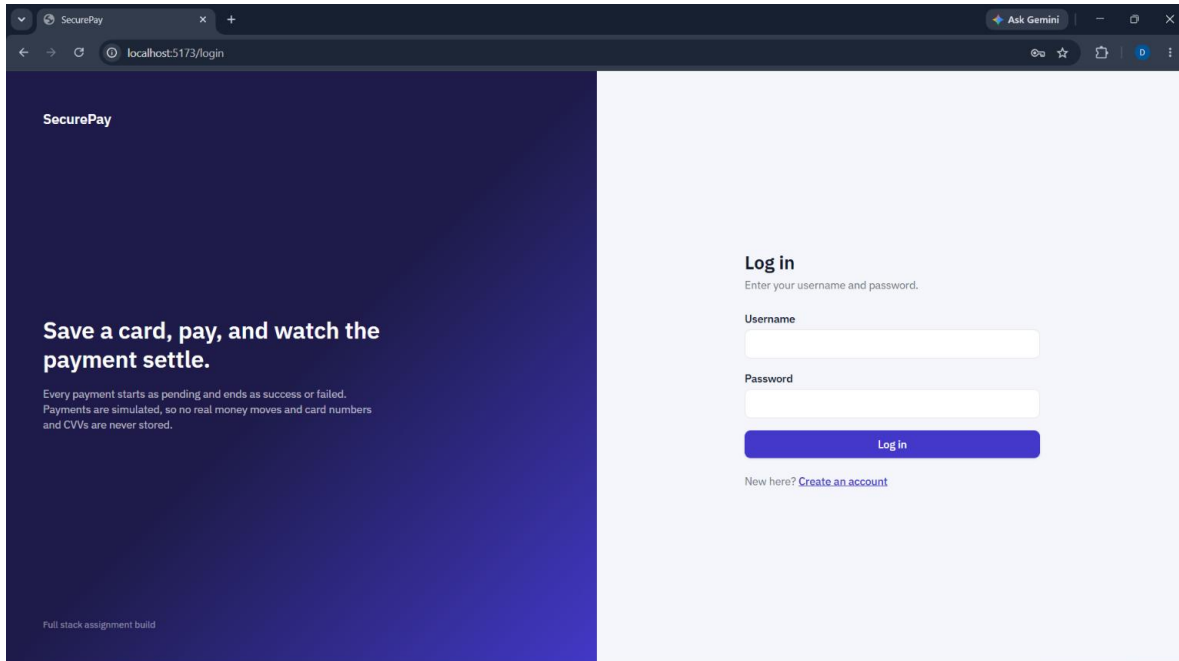
Password

Confirm password

Create account

Already registered? [Log in](#)

Authentication — Login Page



The screenshot shows a web browser window with the URL `localhost:5173/login`. The page is titled "SecurePay" and features a dark blue sidebar on the left with the text "Save a card, pay, and watch the payment settle." and a disclaimer: "Every payment starts as pending and ends as success or failed. Payments are simulated, so no real money moves and card numbers and CVVs are never stored." The main content area is light blue and contains the "Log in" section. It includes a sub-header "Log in" and a note "Enter your username and password." Below this are two input fields: "Username" and "Password". A blue "Log in" button is positioned below the fields. At the bottom of the form, there is a link "New here? Create an account". The footer of the page reads "Full stack assignment build".

SecurePay

Save a card, pay, and watch the payment settle.

Every payment starts as pending and ends as success or failed.
Payments are simulated, so no real money moves and card numbers
and CVVs are never stored.

Full stack assignment build

Log in

Enter your username and password.

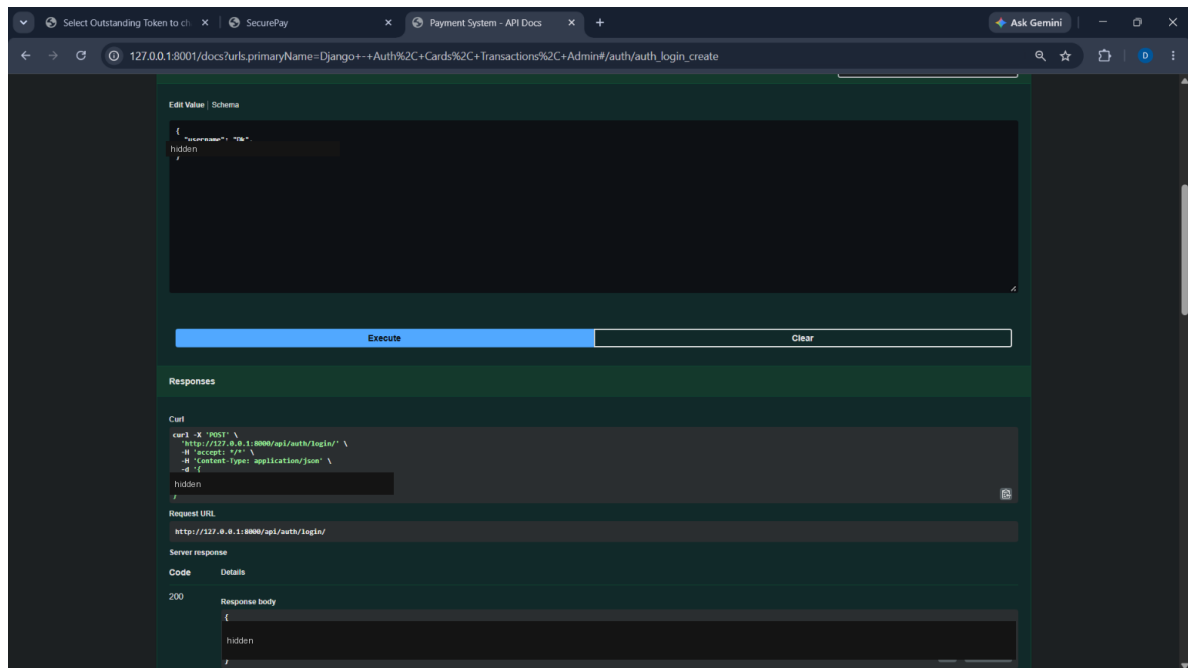
Username

Password

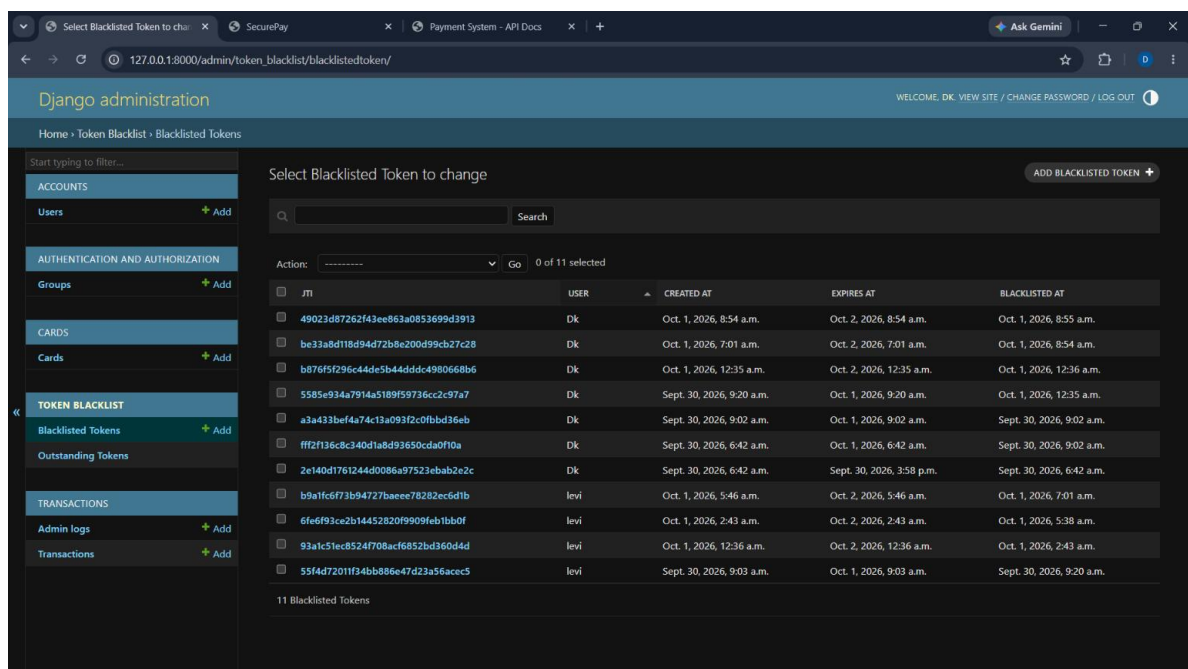
Log in

New here? [Create an account](#)

Authentication — Login API Returns JWT Tokens (secrets hidden)



Authentication — Logout Blacklists Refresh Tokens



Cards — cards Table Has No CVV Column

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

ecommerce_db
fastapi_app_db
payment_db
Tables
Views
Stored Procedures
Functions

Administration Schemas

No object selected

Query 1

```
1. USE payment_db;  
2. DESCRIBE cards;
```

Result Grid

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI		auto_increment
card_holder	varchar(100)	NO			
brand	varchar(20)	NO			
masked_number	varchar(19)	NO			
last4	varchar(4)	NO			
expiry_month	smallint unsigned	NO			
expiry_year	smallint unsigned	NO			

Output

#	Time	Action	Message	Duration / Fetch
1	14:48:30	USE payment_db	0 row(s) affected	0.000 sec
2	14:48:30	DESCRIBE cards	9 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Cards — Stored Card Is Masked (last4 only)

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

ecommerce_db
fastapi_app_db
payment_db
Tables
Views
Stored Procedures
Functions

Administration Schemas

No object selected

Query 1

```
1. USE payment_db;  
2. DESCRIBE cards;  
3. SELECT id, user_id, brand, masked_number, last4, expiry_month, expiry_year FROM cards;
```

Result Grid

id	user_id	brand	masked_number	last4	expiry_month	expiry_year
1	2	VISA	***** 1111	1111	10	2027

Output

#	Time	Action	Message	Duration / Fetch
1	14:48:30	USE payment_db	0 row(s) affected	0.000 sec
2	14:48:30	DESCRIBE cards	9 row(s) returned	0.000 sec / 0.000 sec
3	14:49:05	SELECT id, user_id, brand, masked_number, last4, expiry_month, expiry_year FROM cards U...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Payments — Make Payment (201, SUCCESS)

The screenshot shows a REST client interface with the following details:

- Request Body:**

```
{  "amount": "250.50",  "card_id": 1,  "currency": "INR",  "description": "Order #1001"}
```
- Buttons:** "Execute" (highlighted in blue) and "Clear".
- Responses:**
 - Curl:**

```
curl -X 'POST' \  "http://127.0.0.1:8001/payments" \  -H 'Accept: */*' \  -H 'Authorization: Bearer token hidden' \  -H 'Content-Type: application/json' \  -d '{  "amount": "250.50",  "card_id": 1,  "currency": "INR",  "description": "Order #1001"  }'
```
 - Request URL:** `http://127.0.0.1:8001/payments`
 - Server response:**

Code	Details
201	Response body <pre>{ "reference": "c6ab54a2f6b0d099f107702a0f5d70", "status": "SUCCESS", "amount": "250.50", "currency": "INR", "failure_reason": "", "created_at": "2026-10-01T00:00:00.183134"}</pre>

Payments — Invalid Card Rejected (404)

The screenshot shows a REST client interface with the following details:

- Request Body:**

```
{  "amount": "250.50",  "card_id": 999,  "currency": "INR",  "description": "Order #1001"}
```
- Buttons:** "Execute" (highlighted in blue) and "Clear".
- Responses:**
 - Curl:**

```
curl -X 'POST' \  "http://127.0.0.1:8001/payments" \  -H 'Accept: */*' \  -H 'Authorization: Bearer token hidden' \  -H 'Content-Type: application/json' \  -d '{  "amount": "250.50",  "card_id": 999,  "currency": "INR",  "description": "Order #1001"  }'
```
 - Request URL:** `http://127.0.0.1:8001/payments`
 - Server response:**

Code	Details
404	Error: Not Found Response body <pre>{ "detail": "Card not found"}</pre>

Payments — Dashboard Shows SUCCESS and FAILED

SecurePay

Dashboard

Cards

Make payment

Transactions

Levix
Customer

Log out

Welcome, Levix

Your saved cards and latest payments.

Make a payment

Paid successfully
₹5,525.73

Failed payments
1

Saved cards
1

Recent transactions

Date	Card	Description	Amount	Status
05 Oct 2026, 06:04 pm	**** 1111	Movie Tickets	₹199.99	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Online course	₹4,999.99	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Food delivery	₹75.25	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Mobile Recharge	₹999.99	FAILED Card declined by issuer
05 Oct 2026, 06:04 pm	**** 1111	Order #1001	₹250.50	SUCCESS

Your cards

VISA **** 1111

12/2030

Transactions — Filters: Amount and Date Range

SecurePay

Dashboard

Cards

Make payment

Transactions

Levix
Customer

Log out

Transactions

Filter your payments by date, amount and status.

Status

Success

Min amount

Max amount

From date

dd/mm/yyyy

To date

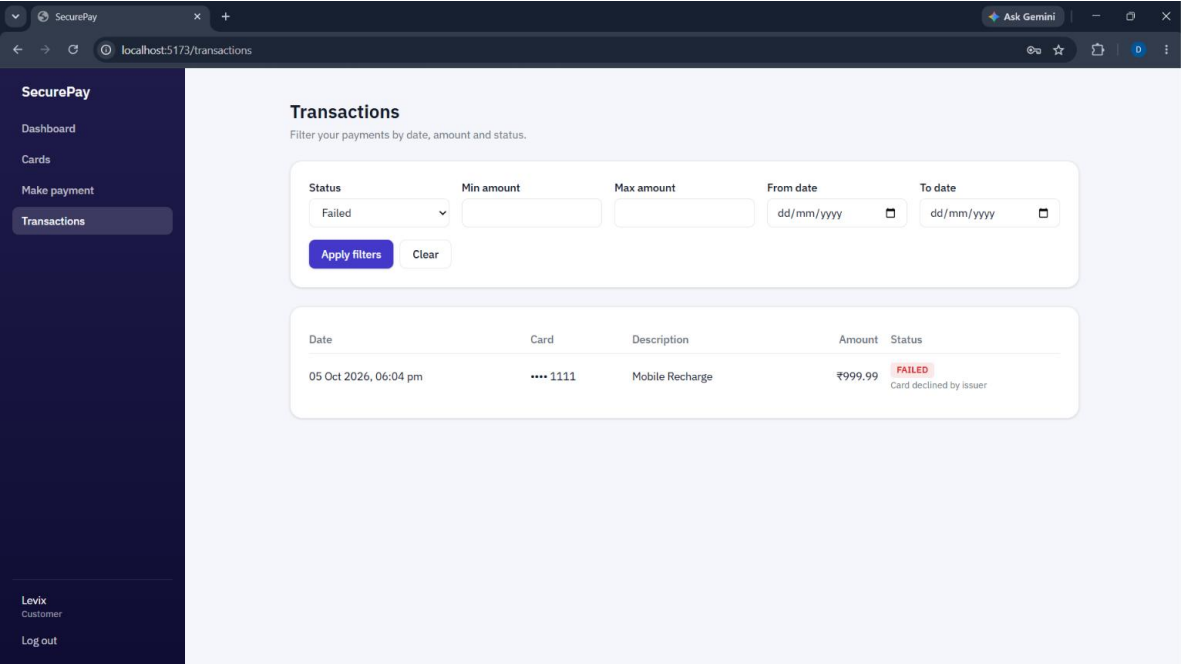
dd/mm/yyyy

Apply filters

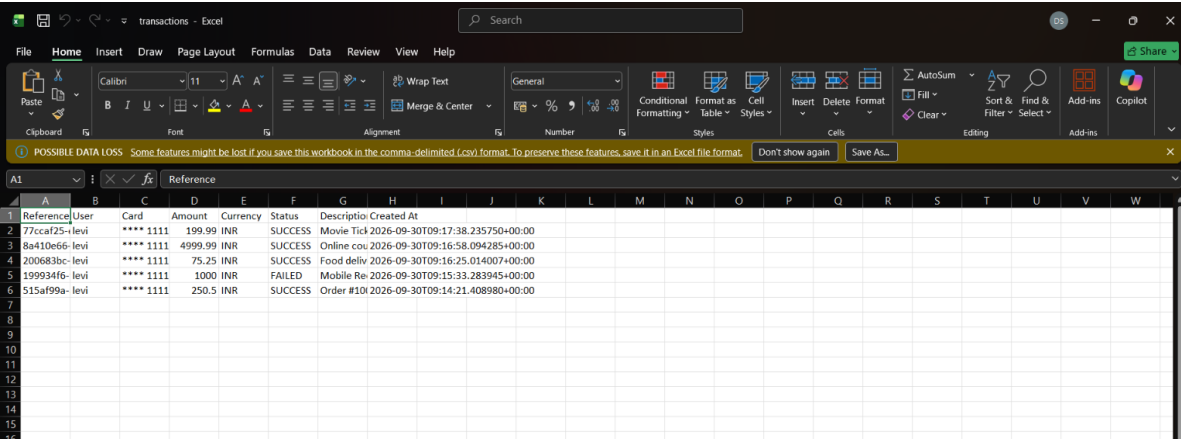
Clear

Date	Card	Description	Amount	Status
05 Oct 2026, 06:04 pm	**** 1111	Movie Tickets	₹199.99	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Online course	₹4,999.99	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Food delivery	₹75.25	SUCCESS
05 Oct 2026, 06:04 pm	**** 1111	Order #1001	₹250.50	SUCCESS

Transactions — Filter by Status (Failed)



Transactions — CSV Export Opened in Excel



Transactions — CSV Export Logged in admin_logs

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

ecommerce_db

fastapi_app_db

payment_db

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

No object selected

Object Info Session

Query 1

1 USE payment_db;

2 SELECT * FROM transactions;

3 SELECT * FROM admin_logs;

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows

id action created_at admin_id

1 Exported transactions to CSV 2026-09-30 09:21:21.858963 1

2 Exported transactions to CSV 2026-10-01 09:43:25.672389 1

admin_logs 2 x

Output

Action Output

Time Action Message Duration / Fetch

1 14:48:06 USE payment_db 0 row(s) affected 0.000 sec

2 14:48:37 SELECT * FROM transactions LIMIT 0, 1000 7 row(s) returned 0.016 sec / 0.000 sec

3 14:49:11 SELECT * FROM admin_logs LIMIT 0, 1000 2 row(s) returned 0.016 sec / 0.000 sec

Admin — Daily Summary

SecurePay

Dashboard

Cards

Make payment

Transactions

Admin

dk Administrator Log out

Admin

Manage users and review cards, transactions and daily payment totals.

Export transactions to CSV

Daily summary Users Cards Transactions

Date	Total	Success	Failed	Pending	Amount collected
2026-10-05	5	4	1	0	₹5,525.73

Admin — Users

SecurePay

Dashboard

Cards

Make payment

Transactions

Admin

dk

Administrator

Log out

Admin

Manage users and review cards, transactions and daily payment totals.

Export transactions to CSV

Daily summary

Users

Cards

Transactions

ID	Username	Email	Role	Joined
2	Levix	levi@gmail.com	Customer	05 Oct 2026, 06:01 pm
1	dk	dineshkumarsenthilk123@gmail.com	Admin	29 Sept 2026, 04:23 pm

Admin — Cards

SecurePay

Dashboard

Cards

Make payment

Transactions

Admin

dk

Administrator

Log out

Admin

Manage users and review cards, transactions and daily payment totals.

Export transactions to CSV

Daily summary

Users

Cards

Transactions

ID	User	Brand	Card	Expiry
1	Levix	VISA	**** * 1111	12/2030

Admin — Transactions

SecurePay

Dashboard

Cards

Make payment

Transactions

Admin

dk

Administrator

Log out

Admin

Manage users and review cards, transactions and daily payment totals.

Export transactions to CSV

Daily summary

Users

Cards

Transactions

Status

All

Date	User	Card	Description	Amount	Status
05 Oct 2026, 06:04 pm	Levix	**** 1111	Movie Tickets	₹199.99	SUCCESS
05 Oct 2026, 06:04 pm	Levix	**** 1111	Online course	₹4,999.99	SUCCESS
05 Oct 2026, 06:04 pm	Levix	**** 1111	Food delivery	₹75.25	SUCCESS
05 Oct 2026, 06:04 pm	Levix	**** 1111	Mobile Recharge	₹999.99	FAILED Card declined by issuer
05 Oct 2026, 06:04 pm	Levix	**** 1111	Order #1001	₹250.50	SUCCESS

Security — Passwords Stored as PBKDF2 Hashes

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SCHEMAS

Filter objects

ecommerce_db

fastapi_app_db

payment_db

Tables

Views

Stored Procedures

Functions

Administration

Schemas

Information

No object selected

Object Info

Session

Query 1

```
1. USE payment_db;
2. DESCRIBE cards;
3. SELECT id, user_id, brand, masked_number, last4, expiry_month, expiry_year FROM cards;
4. SELECT id, username, LEFT(password, 40) AS password_hash, is_staff FROM users;
```

id	username	password_hash	is_staff
1	dk	pbkdf2_sha256\$1000000\$kusID2BG2Ccl3BdE3	1
2	levi	pbkdf2_sha256\$1000000\$ivvd6v75AozfR7Ndy6	0

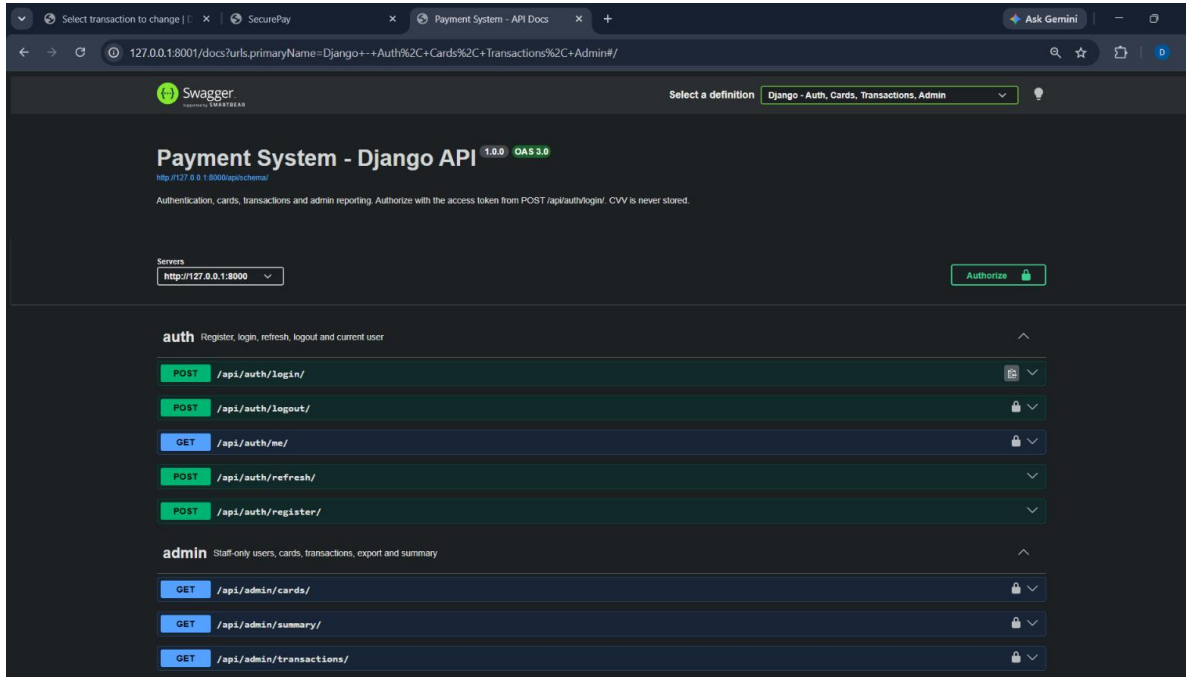
Result 3

#	Time	Action	Message	Duration / Fetch
1	14:48:30	USE payment_db	0 row(s) affected	0.000 sec
2	14:48:30	DESCRIBE cards	9 row(s) returned	0.000 sec / 0.000 sec
3	14:49:05	SELECT id, user_id, brand, masked_number, last4, expiry_month, expiry_year FROM cards LIMIT 1	1 row(s) returned	0.000 sec / 0.000 sec
4	14:49:29	SELECT id, username, LEFT(password, 40) AS password_hash, is_staff FROM users LIMIT 0...	2 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Swagger — Django Auth and Admin Endpoints



The image shows the Swagger UI for the "Payment System - Django API". The interface is dark-themed. At the top, there's a header with the Swagger logo and the API title. Below the header, there's a section for "auth" with endpoints for login, logout, me, refresh, and register. The "admin" section follows, listing endpoints for cards, summary, and transactions. Each endpoint is represented by a colored button (green for POST, blue for GET) and a lock icon indicating authentication requirements.

Payment System - Django API 1.0.0 OAS 3.0

Authentication, cards, transactions and admin reporting. Authorize with the access token from POST /api/auth/login/. CVV is never stored.

Servers: http://127.0.0.1:8000

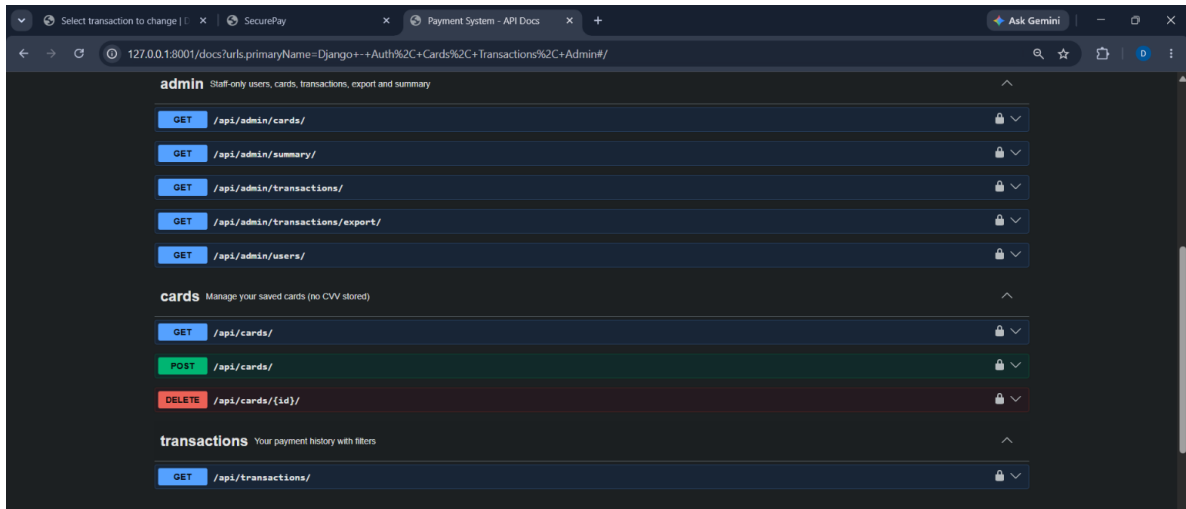
auth Register, login, refresh, logout and current user

- POST /api/auth/login/
- POST /api/auth/logout/
- GET /api/auth/me/
- POST /api/auth/refresh/
- POST /api/auth/register/

admin Staff-only users, cards, transactions, export and summary

- GET /api/admin/cards/
- GET /api/admin/summary/
- GET /api/admin/transactions/

Swagger — Django Admin, Cards and Transactions Endpoints



This image shows a different view of the Swagger UI, focusing on the "admin", "cards", and "transactions" sections. The "admin" section lists endpoints for cards, summary, transactions, export, and users. The "cards" section shows endpoints for listing, creating, and deleting cards. The "transactions" section shows an endpoint for listing transactions. Each endpoint is represented by a colored button (blue for GET, green for POST, red for DELETE) and a lock icon.

admin Staff-only users, cards, transactions, export and summary

- GET /api/admin/cards/
- GET /api/admin/summary/
- GET /api/admin/transactions/
- GET /api/admin/transactions/export/
- GET /api/admin/users/

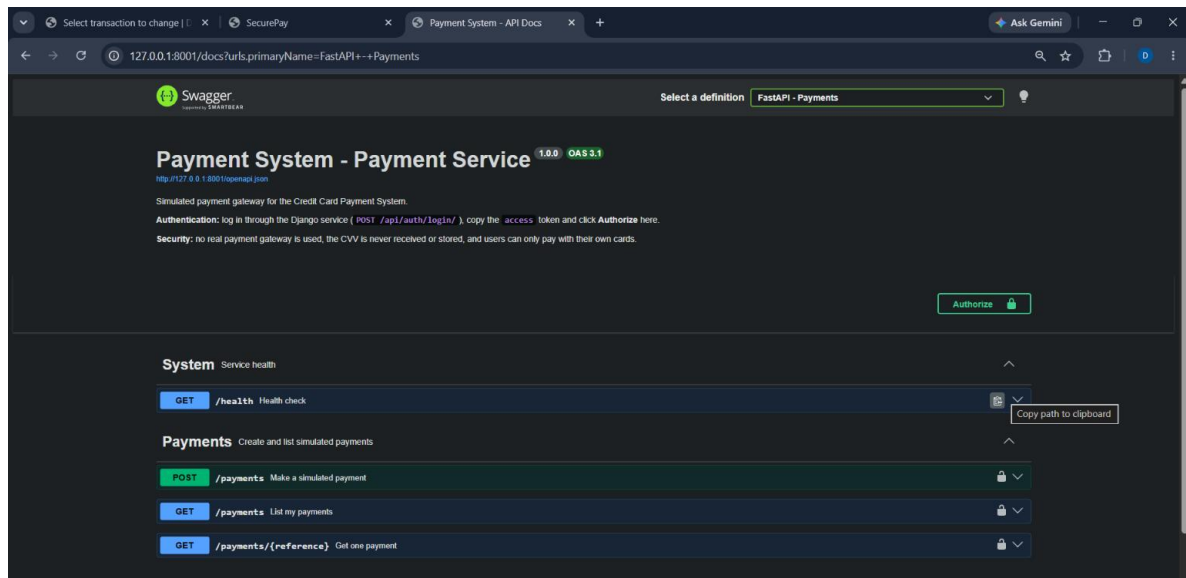
cards Manage your saved cards (no CVV stored)

- GET /api/cards/
- POST /api/cards/
- DELETE /api/cards/{id}/

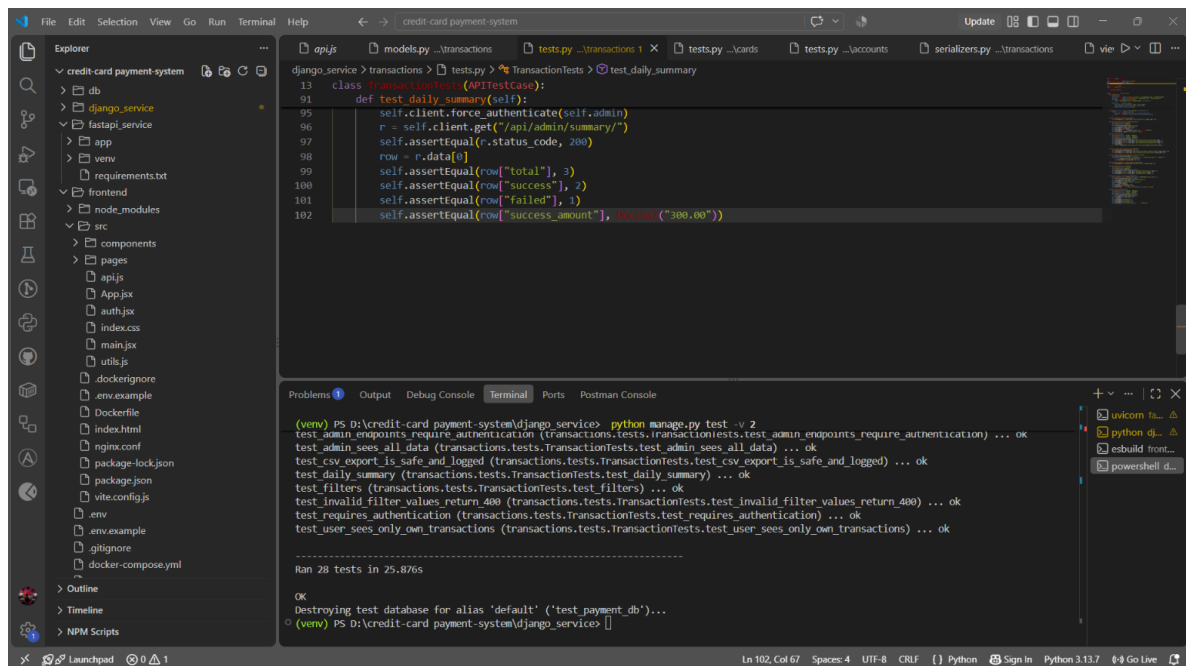
transactions Your payment history with filters

- GET /api/transactions/

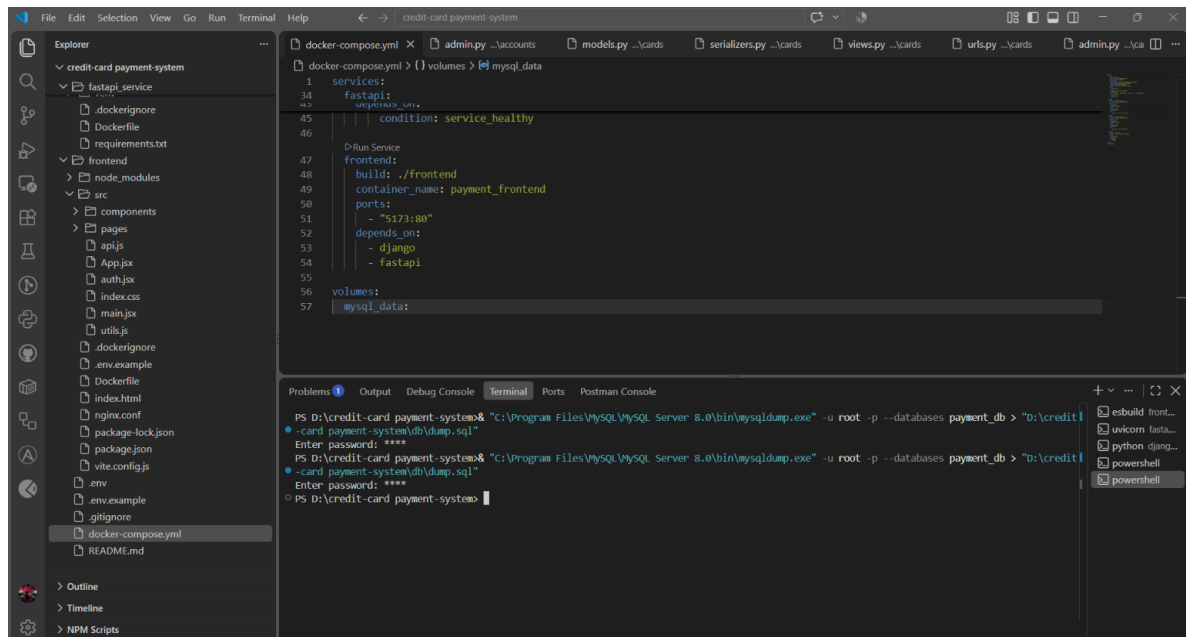
Swagger — FastAPI Payments Endpoints



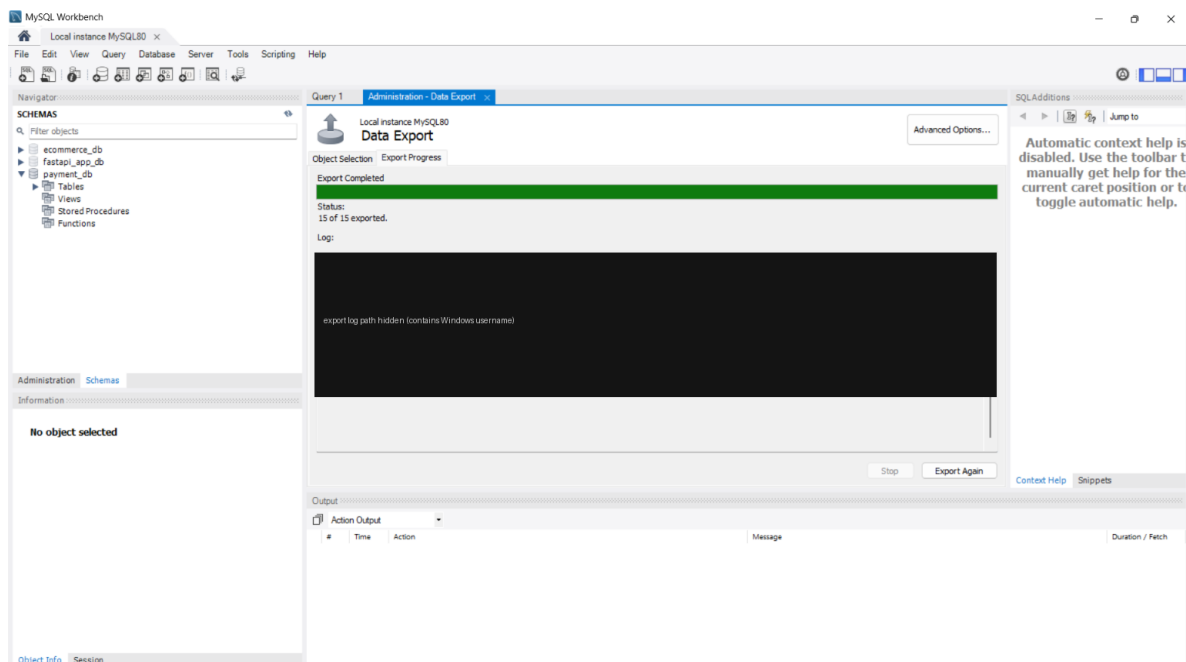
Testing — Django Tests (28 Passed)



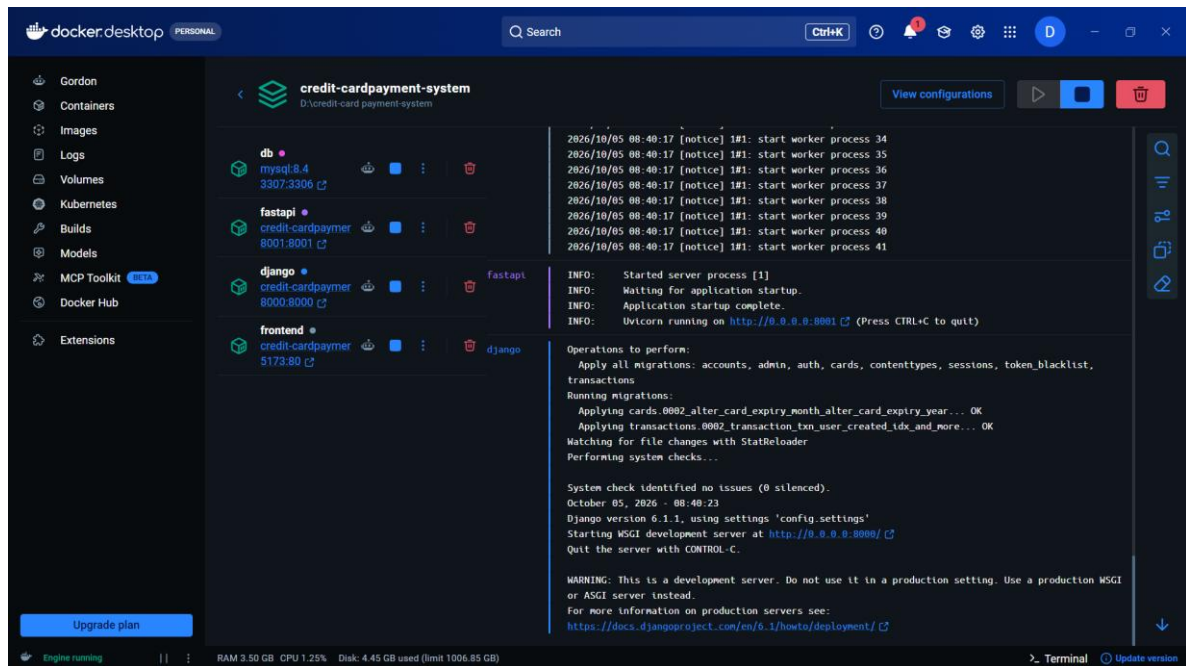
Database — mysqldump Command and docker-compose.yml



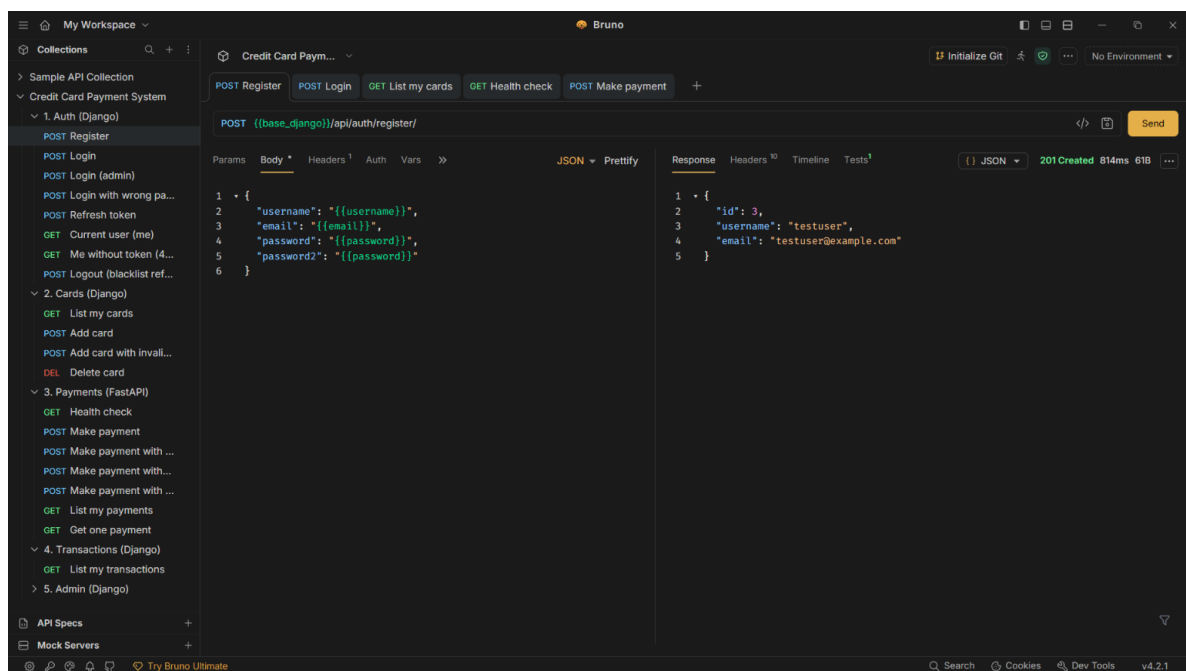
Database — MySQL Workbench Export Completed



Docker — Containers Running



Postman Collection — Imported with All Folders, Register Returns 201



Postman Collection — Login Returns 200, 2 Tests Passed (tokens hidden)

